LithoVault Yellow Paper

Autonomous Treasury Infrastructure for Lithosphere

Enterprise-grade vaults, governed AI execution, DNNS identity, APP payment rails, and MultX cross-chain treasury settlement

Version 0.9 draft — March 30, 2026

Reference network: Lithosphere Makalu — Chain ID 700777 — RPC https://rpc.litho.ai

Product	LithoVault
Positioning	Autonomous CFO and treasury operating system
Core dependencies	Lithosphere, DNNS, MultX, APP, Quantt pipelines
Target users	Enterprises, DAOs, treasury teams, professional on-chain allocators
Repository deliverables	Web app, mobile app, API, SDKs, CLI, contracts, deployment manifests

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1. Executive Summary

1. Executive Summary presents the reference design for LithoVault within a governed enterprise treasury context. The discussion assumes a deployment on Lithosphere Makalu and uses APP, DNNS, and MultX-compatible interfaces as foundational coordination layers.

Vision

LithoVault treats treasury management as a controlled execution problem in which intent, policy, simulation, authorization, and settlement are separate stages that can be logged and reviewed independently. The platform is designed so that AI systems never become unbounded operators; every recommendation is expressed as a typed treasury intent, scored by confidence, constrained by policy, and either executed under delegated limits or escalated to governance. By separating vault state from strategy logic and by separating strategy logic from bridge and protocol adapters, the system can evolve individual components without invalidating accounting or governance assumptions. A treasury deployment can map each business unit, legal entity, or mandate to an isolated vault with its own policy envelope, signer set, asset list, reporting cadence, and permissible protocol adapters. The intended operational benefit is not merely higher yield but tighter control over how liquidity moves, why it moves, and what evidence remains after the fact.

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- Execution adapters must be replaceable and versioned to avoid coupling treasury accounting to any single DeFi venue or bridge implementation.

Control	Objective	Reference behavior
Policy envelope	Bound autonomous actions	Reject moves outside board-approved thresholds
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System thesis

LithoVault treats treasury management as a controlled execution problem in which intent, policy, simulation, authorization, and settlement are separate stages that can be logged and reviewed independently. The platform is designed so that AI systems never become unbounded operators; every recommendation is expressed as a typed treasury intent, scored by confidence, constrained by policy, and either executed under delegated limits or escalated to governance. By separating vault state from strategy logic and by separating strategy logic from bridge and protocol adapters, the system can evolve individual components without invalidating accounting or governance assumptions. A treasury deployment can map each business unit, legal entity, or mandate to an isolated vault with its own policy envelope, signer set, asset list, reporting cadence, and

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Deliverables

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2. Problem Statement

2. Problem Statement presents the reference design for LithoVault within a governed enterprise treasury context. The discussion assumes a deployment on Lithosphere Makalu and uses APP, DNNS, and MultX-compatible interfaces as foundational coordination layers.

Operational pain

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Policy envelope	Bound autonomous actions	Reject moves outside board-approved thresholds
Simulation	Pre-trade validation	Estimate route, slippage, and failure modes
Attestation log	Auditability	Persist model vote, signer approval, and transaction hash

Why existing tools fall short

LithoVault treats treasury management as a controlled execution problem in which intent, policy, simulation, authorization, and settlement are separate stages that can be logged and reviewed independently. The platform is designed so that AI systems never become unbounded operators; every recommendation is expressed as a typed treasury intent, scored by confidence, constrained by policy, and either executed under delegated limits or escalated to governance. By separating vault state from strategy logic and by separating strategy logic from bridge and protocol adapters, the system can evolve individual components without invalidating accounting or governance assumptions. A treasury deployment can map each business unit, legal entity, or mandate to an isolated vault with its own policy envelope, signer set, asset list, reporting cadence, and permissible protocol adapters. The intended operational benefit is not merely higher yield but tighter control over how liquidity moves, why it moves, and what evidence remains after the fact.

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- Policy evaluation precedes execution and determines whether an action can be delegated, queued, rate-limited, or rejected.
- Every material state transition emits a structured event so that operators can reconstruct what happened without depending on off-chain memory.
- Execution adapters must be replaceable and versioned to avoid coupling treasury accounting to any single DeFi venue or bridge implementation.

Control	Objective	Reference behavior
Policy envelope	Bound autonomous actions	Reject moves outside board-approved thresholds
Simulation	Pre-trade validation	Estimate route, slippage, and failure modes
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3. Design Principles

3. Design Principles presents the reference design for LithoVault within a governed enterprise treasury context. The discussion assumes a deployment on Lithosphere Makalu and uses APP, DNNS, and MultX-compatible interfaces as foundational coordination layers.

Determinism

LithoVault treats treasury management as a controlled execution problem in which intent, policy, simulation, authorization, and settlement are separate stages that can be logged and reviewed independently. The platform is designed so that AI systems never become unbounded operators; every recommendation is expressed as a typed treasury intent, scored by confidence, constrained by policy, and either executed under delegated limits or escalated to governance. By separating vault state from strategy logic and by separating strategy logic from bridge and protocol adapters, the system can evolve individual components without invalidating accounting or governance assumptions. A treasury deployment can map each business unit, legal entity, or mandate to an isolated vault with its own policy envelope, signer set, asset list, reporting cadence, and permissible protocol adapters. The intended operational benefit is not merely higher yield but tighter control over how liquidity moves, why it moves, and what evidence remains after the fact.

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Composability

LithoVault treats treasury management as a controlled execution problem in which intent, policy, simulation, authorization, and settlement are separate stages that can be logged and reviewed independently. The platform is designed so that AI systems never become unbounded operators; every recommendation is expressed as a typed treasury intent, scored by confidence, constrained by policy, and either executed under delegated limits or escalated to governance. By separating vault state from strategy logic and by separating strategy logic from bridge and protocol adapters, the system can evolve individual components without invalidating accounting or governance assumptions. A treasury deployment can map each business unit, legal entity, or mandate to an isolated vault with its own policy envelope, signer set, asset list, reporting cadence, and

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Human override

LithoVault treats treasury management as a controlled execution problem in which intent, policy, simulation, authorization, and settlement are separate stages that can be logged and reviewed independently. The platform is designed so that AI systems never become unbounded operators; every recommendation is expressed as a typed treasury intent, scored by confidence, constrained by policy, and either executed under delegated limits or escalated to governance. By separating vault state from strategy logic and by separating strategy logic from bridge and protocol adapters, the system can evolve individual components without invalidating accounting or governance assumptions. A treasury deployment can map each business unit, legal entity, or mandate to an isolated vault with its own policy envelope, signer set, asset list, reporting cadence, and permissible protocol adapters. The intended operational benefit is not merely higher yield but tighter control over how liquidity moves, why it moves, and what evidence remains after the fact.

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4. System Architecture

4. System Architecture presents the reference design for LithoVault within a governed enterprise treasury context. The discussion assumes a deployment on Lithosphere Makalu and uses APP, DNNS, and MultX-compatible interfaces as foundational coordination layers.

Control plane

LithoVault treats treasury management as a controlled execution problem in which intent, policy, simulation, authorization, and settlement are separate stages that can be logged and reviewed independently. The platform is designed so that AI systems never become unbounded operators; every recommendation is expressed as a typed treasury intent, scored by confidence, constrained by policy, and either executed under delegated limits or escalated to governance. By separating vault state from strategy logic and by separating strategy logic from bridge and protocol adapters, the system can evolve individual components without invalidating accounting or governance assumptions. A treasury deployment can map each business unit, legal entity, or mandate to an isolated vault with its own policy envelope, signer set, asset list, reporting cadence, and permissible protocol adapters. The intended operational benefit is not merely higher yield but tighter control over how liquidity moves, why it moves, and what evidence remains after the fact.

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Attestation log	Auditability	Persist model vote, signer approval, and transaction hash

Execution plane

LithoVault treats treasury management as a controlled execution problem in which intent, policy, simulation, authorization, and settlement are separate stages that can be logged and reviewed independently. The platform is designed so that AI systems never become unbounded operators; every recommendation is expressed as a typed treasury intent, scored by confidence, constrained by policy, and either executed under delegated limits or escalated to governance. By separating vault state from strategy logic and by separating strategy logic from bridge and protocol adapters, the system can evolve individual components without invalidating accounting or governance assumptions. A treasury deployment can map each business unit, legal entity, or mandate to an isolated vault with its own policy envelope, signer set, asset list, reporting cadence, and

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- Every material state transition emits a structured event so that operators can reconstruct what happened without depending on off-chain memory.
- Execution adapters must be replaceable and versioned to avoid coupling treasury accounting to any single DeFi venue or bridge implementation.

Control	Objective	Reference behavior
Policy envelope	Bound autonomous actions	Reject moves outside board-approved thresholds
Simulation	Pre-trade validation	Estimate route, slippage, and failure modes
Attestation log	Auditability	Persist model vote, signer approval, and transaction hash

Observability plane

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- Execution adapters must be replaceable and versioned to avoid coupling treasury accounting to any single DeFi venue or bridge implementation.

Control	Objective	Reference behavior
Policy envelope	Bound autonomous actions	Reject moves outside board-approved thresholds
Simulation	Pre-trade validation	Estimate route, slippage, and failure modes
Attestation log	Auditability	Persist model vote, signer approval, and transaction hash

5. Vault Model

5. Vault Model presents the reference design for LithoVault within a governed enterprise treasury context. The discussion assumes a deployment on Lithosphere Makalu and uses APP, DNNS, and MultX-compatible interfaces as foundational coordination layers.

Balances

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Attestation log	Auditability	Persist model vote, signer approval, and transaction hash

Policies

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Simulation	Pre-trade validation	Estimate route, slippage, and failure modes
Attestation log	Auditability	Persist model vote, signer approval, and transaction hash

Jobs and queues

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Simulation	Pre-trade validation	Estimate route, slippage, and failure modes
Attestation log	Auditability	Persist model vote, signer approval, and transaction hash

6. Governance and Roles

6. Governance and Roles presents the reference design for LithoVault within a governed enterprise treasury context. The discussion assumes a deployment on Lithosphere Makalu and uses APP, DNNS, and MultX-compatible interfaces as foundational coordination layers.

Signer groups

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- Execution adapters must be replaceable and versioned to avoid coupling treasury accounting to any single DeFi venue or bridge implementation.

Control	Objective	Reference behavior
Policy envelope	Bound autonomous actions	Reject moves outside board-approved thresholds
Simulation	Pre-trade validation	Estimate route, slippage, and failure modes
Attestation log	Auditability	Persist model vote, signer approval, and transaction hash

Delegated limits

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Control	Objective	Reference behavior
Policy envelope	Bound autonomous actions	Reject moves outside board-approved thresholds
Simulation	Pre-trade validation	Estimate route, slippage, and failure modes
Attestation log	Auditability	Persist model vote, signer approval, and transaction hash

Emergency controls

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Control	Objective	Reference behavior
Policy envelope	Bound autonomous actions	Reject moves outside board-approved thresholds
Simulation	Pre-trade validation	Estimate route, slippage, and failure modes
Attestation log	Auditability	Persist model vote, signer approval, and transaction hash

7. APP Payment Rail Integration

7. APP Payment Rail Integration presents the reference design for LithoVault within a governed enterprise treasury context. The discussion assumes a deployment on Lithosphere Makalu and uses APP, DNNS, and MultX-compatible interfaces as foundational coordination layers.

Intent accounting

LithoVault treats treasury management as a controlled execution problem in which intent, policy, simulation, authorization, and settlement are separate stages that can be logged and reviewed independently. The platform is designed so that AI systems never become unbounded operators; every recommendation is expressed as a typed treasury intent, scored by confidence, constrained by policy, and either executed under delegated limits or escalated to governance. By separating vault state from strategy logic and by separating strategy logic from bridge and protocol adapters, the system can evolve individual components without invalidating accounting or governance assumptions. A treasury deployment can map each business unit, legal entity, or mandate to an isolated vault with its own policy envelope, signer set, asset list, reporting cadence, and permissible protocol adapters. The intended operational benefit is not merely higher yield but tighter control over how liquidity moves, why it moves, and what evidence remains after the fact.

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Simulation	Pre-trade validation	Estimate route, slippage, and failure modes
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Payment jobs

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Attestation log	Auditability	Persist model vote, signer approval, and transaction hash

Attribution

LithoVault treats treasury management as a controlled execution problem in which intent, policy, simulation, authorization, and settlement are separate stages that can be logged and reviewed independently. The platform is designed so that AI systems never become unbounded operators; every recommendation is expressed as a typed treasury intent, scored by confidence, constrained by policy, and either executed under delegated limits or escalated to governance. By separating vault state from strategy logic and by separating strategy logic from bridge and protocol adapters, the system can evolve individual components without invalidating accounting or governance assumptions. A treasury deployment can map each business unit, legal entity, or mandate to an isolated vault with its own policy envelope, signer set, asset list, reporting cadence, and permissible protocol adapters. The intended operational benefit is not merely higher yield but tighter control over how liquidity moves, why it moves, and what evidence remains after the fact.

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- Policy evaluation precedes execution and determines whether an action can be delegated, queued, rate-limited, or rejected.
- Every material state transition emits a structured event so that operators can reconstruct what happened without depending on off-chain memory.
- Execution adapters must be replaceable and versioned to avoid coupling treasury accounting to any single DeFi venue or bridge implementation.

Control	Objective	Reference behavior
Policy envelope	Bound autonomous actions	Reject moves outside board-approved thresholds
Simulation	Pre-trade validation	Estimate route, slippage, and failure modes
Attestation log	Auditability	Persist model vote, signer approval, and transaction hash

8. DNNS Identity Model

8. DNNS Identity Model presents the reference design for LithoVault within a governed enterprise treasury context. The discussion assumes a deployment on Lithosphere Makalu and uses APP, DNNS, and MultX-compatible interfaces as foundational coordination layers.

Agent identities

LithoVault treats treasury management as a controlled execution problem in which intent, policy, simulation, authorization, and settlement are separate stages that can be logged and reviewed independently. The platform is designed so that AI systems never become unbounded operators; every recommendation is expressed as a typed treasury intent, scored by confidence, constrained by policy, and either executed under delegated limits or escalated to governance. By separating vault state from strategy logic and by separating strategy logic from bridge and protocol adapters, the system can evolve individual components without invalidating accounting or governance assumptions. A treasury deployment can map each business unit, legal entity, or mandate to an isolated vault with its own policy envelope, signer set, asset list, reporting cadence, and permissible protocol adapters. The intended operational benefit is not merely higher yield but tighter control over how liquidity moves, why it moves, and what evidence remains after the fact.

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Control	Objective	Reference behavior
Policy envelope	Bound autonomous actions	Reject moves outside board-approved thresholds
Simulation	Pre-trade validation	Estimate route, slippage, and failure modes
Attestation log	Auditability	Persist model vote, signer approval, and transaction hash

Vault namespaces

LithoVault treats treasury management as a controlled execution problem in which intent, policy, simulation, authorization, and settlement are separate stages that can be logged and reviewed independently. The platform is designed so that AI systems never become unbounded operators; every recommendation is expressed as a typed treasury intent, scored by confidence, constrained by policy, and either executed under delegated limits or escalated to governance. By separating vault state from strategy logic and by separating strategy logic from bridge and protocol adapters, the system can evolve individual components without invalidating accounting or governance assumptions. A treasury deployment can map each business unit, legal entity, or mandate to an isolated vault with its own policy envelope, signer set, asset list, reporting cadence, and

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Simulation	Pre-trade validation	Estimate route, slippage, and failure modes
Attestation log	Auditability	Persist model vote, signer approval, and transaction hash

Controller rotation

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Control	Objective	Reference behavior
Policy envelope	Bound autonomous actions	Reject moves outside board-approved thresholds
Simulation	Pre-trade validation	Estimate route, slippage, and failure modes
Attestation log	Auditability	Persist model vote, signer approval, and transaction hash

9. Quantt Intelligence Layer

9. Quantt Intelligence Layer presents the reference design for LithoVault within a governed enterprise treasury context. The discussion assumes a deployment on Lithosphere Makalu and uses APP, DNNS, and MultX-compatible interfaces as foundational coordination layers.

Signal ingestion

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Attestation log	Auditability	Persist model vote, signer approval, and transaction hash

Consensus logic

LithoVault treats treasury management as a controlled execution problem in which intent, policy, simulation, authorization, and settlement are separate stages that can be logged and reviewed independently. The platform is designed so that AI systems never become unbounded operators; every recommendation is expressed as a typed treasury intent, scored by confidence, constrained by policy, and either executed under delegated limits or escalated to governance. By separating vault state from strategy logic and by separating strategy logic from bridge and protocol adapters, the system can evolve individual components without invalidating accounting or governance assumptions. A treasury deployment can map each business unit, legal entity, or mandate to an isolated vault with its own policy envelope, signer set, asset list, reporting cadence, and

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- Policy evaluation precedes execution and determines whether an action can be delegated, queued, rate-limited, or rejected.
- Every material state transition emits a structured event so that operators can reconstruct what happened without depending on off-chain memory.
- Execution adapters must be replaceable and versioned to avoid coupling treasury accounting to any single DeFi venue or bridge implementation.

Control	Objective	Reference behavior
Policy envelope	Bound autonomous actions	Reject moves outside board-approved thresholds
Simulation	Pre-trade validation	Estimate route, slippage, and failure modes
Attestation log	Auditability	Persist model vote, signer approval, and transaction hash

Decision export

LithoVault treats treasury management as a controlled execution problem in which intent, policy, simulation, authorization, and settlement are separate stages that can be logged and reviewed independently. The platform is designed so that AI systems never become unbounded operators; every recommendation is expressed as a typed treasury intent, scored by confidence, constrained by policy, and either executed under delegated limits or escalated to governance. By separating vault state from strategy logic and by separating strategy logic from bridge and protocol adapters, the system can evolve individual components without invalidating accounting or governance assumptions. A treasury deployment can map each business unit, legal entity, or mandate to an isolated vault with its own policy envelope, signer set, asset list, reporting cadence, and permissible protocol adapters. The intended operational benefit is not merely higher yield but tighter control over how liquidity moves, why it moves, and what evidence remains after the fact.

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Control	Objective	Reference behavior
Policy envelope	Bound autonomous actions	Reject moves outside board-approved thresholds
Simulation	Pre-trade validation	Estimate route, slippage, and failure modes
Attestation log	Auditability	Persist model vote, signer approval, and transaction hash

10. Strategy Engine

10. Strategy Engine presents the reference design for LithoVault within a governed enterprise treasury context. The discussion assumes a deployment on Lithosphere Makalu and uses APP, DNNS, and MultX-compatible interfaces as foundational coordination layers.

Yield

LithoVault treats treasury management as a controlled execution problem in which intent, policy, simulation, authorization, and settlement are separate stages that can be logged and reviewed independently. The platform is designed so that AI systems never become unbounded operators; every recommendation is expressed as a typed treasury intent, scored by confidence, constrained by policy, and either executed under delegated limits or escalated to governance. By separating vault state from strategy logic and by separating strategy logic from bridge and protocol adapters, the system can evolve individual components without invalidating accounting or governance assumptions. A treasury deployment can map each business unit, legal entity, or mandate to an isolated vault with its own policy envelope, signer set, asset list, reporting cadence, and permissible protocol adapters. The intended operational benefit is not merely higher yield but tighter control over how liquidity moves, why it moves, and what evidence remains after the fact.

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Attestation log	Auditability	Persist model vote, signer approval, and transaction hash

Rebalancing

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Attestation log	Auditability	Persist model vote, signer approval, and transaction hash

Risk reduction

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Attestation log	Auditability	Persist model vote, signer approval, and transaction hash

11. DeFi Adapter Layer

11. DeFi Adapter Layer presents the reference design for LithoVault within a governed enterprise treasury context. The discussion assumes a deployment on Lithosphere Makalu and uses APP, DNNS, and MultX-compatible interfaces as foundational coordination layers.

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- Policy evaluation precedes execution and determines whether an action can be delegated, queued, rate-limited, or rejected.
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- Execution adapters must be replaceable and versioned to avoid coupling treasury accounting to any single DeFi venue or bridge implementation.

Control	Objective	Reference behavior
Policy envelope	Bound autonomous actions	Reject moves outside board-approved thresholds
Simulation	Pre-trade validation	Estimate route, slippage, and failure modes
Attestation log	Auditability	Persist model vote, signer approval, and transaction hash

Stablecoin operations

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Policy envelope	Bound autonomous actions	Reject moves outside board-approved thresholds
Simulation	Pre-trade validation	Estimate route, slippage, and failure modes
Attestation log	Auditability	Persist model vote, signer approval, and transaction hash

AMM positions

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Policy envelope	Bound autonomous actions	Reject moves outside board-approved thresholds
Simulation	Pre-trade validation	Estimate route, slippage, and failure modes
Attestation log	Auditability	Persist model vote, signer approval, and transaction hash

12. MultX Cross-Chain Coordination

12. MultX Cross-Chain Coordination presents the reference design for LithoVault within a governed enterprise treasury context. The discussion assumes a deployment on Lithosphere Makalu and uses APP, DNNS, and MultX-compatible interfaces as foundational coordination layers.

Bridge intents

LithoVault treats treasury management as a controlled execution problem in which intent, policy, simulation, authorization, and settlement are separate stages that can be logged and reviewed independently. The platform is designed so that AI systems never become unbounded operators; every recommendation is expressed as a typed treasury intent, scored by confidence, constrained by policy, and either executed under delegated limits or escalated to governance. By separating vault state from strategy logic and by separating strategy logic from bridge and protocol adapters, the system can evolve individual components without invalidating accounting or governance assumptions. A treasury deployment can map each business unit, legal entity, or mandate to an isolated vault with its own policy envelope, signer set, asset list, reporting cadence, and permissible protocol adapters. The intended operational benefit is not merely higher yield but tighter control over how liquidity moves, why it moves, and what evidence remains after the fact.

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Simulation	Pre-trade validation	Estimate route, slippage, and failure modes
Attestation log	Auditability	Persist model vote, signer approval, and transaction hash

Validator attestations

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Attestation log	Auditability	Persist model vote, signer approval, and transaction hash

Finalization

LithoVault treats treasury management as a controlled execution problem in which intent, policy, simulation, authorization, and settlement are separate stages that can be logged and reviewed independently. The platform is designed so that AI systems never become unbounded operators; every recommendation is expressed as a typed treasury intent, scored by confidence, constrained by policy, and either executed under delegated limits or escalated to governance. By separating vault state from strategy logic and by separating strategy logic from bridge and protocol adapters, the system can evolve individual components without invalidating accounting or governance assumptions. A treasury deployment can map each business unit, legal entity, or mandate to an isolated vault with its own policy envelope, signer set, asset list, reporting cadence, and permissible protocol adapters. The intended operational benefit is not merely higher yield but tighter control over how liquidity moves, why it moves, and what evidence remains after the fact.

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- Policy evaluation precedes execution and determines whether an action can be delegated, queued, rate-limited, or rejected.
- Every material state transition emits a structured event so that operators can reconstruct what happened without depending on off-chain memory.
- Execution adapters must be replaceable and versioned to avoid coupling treasury accounting to any single DeFi venue or bridge implementation.

Control	Objective	Reference behavior
Policy envelope	Bound autonomous actions	Reject moves outside board-approved thresholds
Simulation	Pre-trade validation	Estimate route, slippage, and failure modes
Attestation log	Auditability	Persist model vote, signer approval, and transaction hash

13. ZK Identity Extension

13. ZK Identity Extension presents the reference design for LithoVault within a governed enterprise treasury context. The discussion assumes a deployment on Lithosphere Makalu and uses APP, DNNS, and MultX-compatible interfaces as foundational coordination layers.

Commitments

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Control	Objective	Reference behavior
Policy envelope	Bound autonomous actions	Reject moves outside board-approved thresholds
Simulation	Pre-trade validation	Estimate route, slippage, and failure modes
Attestation log	Auditability	Persist model vote, signer approval, and transaction hash

Membership proofs

LithoVault treats treasury management as a controlled execution problem in which intent, policy, simulation, authorization, and settlement are separate stages that can be logged and reviewed independently. The platform is designed so that AI systems never become unbounded operators; every recommendation is expressed as a typed treasury intent, scored by confidence, constrained by policy, and either executed under delegated limits or escalated to governance. By separating vault state from strategy logic and by separating strategy logic from bridge and protocol adapters, the system can evolve individual components without invalidating accounting or governance assumptions. A treasury deployment can map each business unit, legal entity, or mandate to an isolated vault with its own policy envelope, signer set, asset list, reporting cadence, and

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Simulation	Pre-trade validation	Estimate route, slippage, and failure modes
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Nullifiers

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Control	Objective	Reference behavior
Policy envelope	Bound autonomous actions	Reject moves outside board-approved thresholds
Simulation	Pre-trade validation	Estimate route, slippage, and failure modes
Attestation log	Auditability	Persist model vote, signer approval, and transaction hash

14. Risk Framework

14. Risk Framework presents the reference design for LithoVault within a governed enterprise treasury context. The discussion assumes a deployment on Lithosphere Makalu and uses APP, DNNS, and MultX-compatible interfaces as foundational coordination layers.

Concentration limits

LithoVault treats treasury management as a controlled execution problem in which intent, policy, simulation, authorization, and settlement are separate stages that can be logged and reviewed independently. The platform is designed so that AI systems never become unbounded operators; every recommendation is expressed as a typed treasury intent, scored by confidence, constrained by policy, and either executed under delegated limits or escalated to governance. By separating vault state from strategy logic and by separating strategy logic from bridge and protocol adapters, the system can evolve individual components without invalidating accounting or governance assumptions. A treasury deployment can map each business unit, legal entity, or mandate to an isolated vault with its own policy envelope, signer set, asset list, reporting cadence, and permissible protocol adapters. The intended operational benefit is not merely higher yield but tighter control over how liquidity moves, why it moves, and what evidence remains after the fact.

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Attestation log	Auditability	Persist model vote, signer approval, and transaction hash

Bridge caps

LithoVault treats treasury management as a controlled execution problem in which intent, policy, simulation, authorization, and settlement are separate stages that can be logged and reviewed independently. The platform is designed so that AI systems never become unbounded operators; every recommendation is expressed as a typed treasury intent, scored by confidence, constrained by policy, and either executed under delegated limits or escalated to governance. By separating vault state from strategy logic and by separating strategy logic from bridge and protocol adapters, the system can evolve individual components without invalidating accounting or governance assumptions. A treasury deployment can map each business unit, legal entity, or mandate to an isolated vault with its own policy envelope, signer set, asset list, reporting cadence, and

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- Policy evaluation precedes execution and determines whether an action can be delegated, queued, rate-limited, or rejected.
- Every material state transition emits a structured event so that operators can reconstruct what happened without depending on off-chain memory.
- Execution adapters must be replaceable and versioned to avoid coupling treasury accounting to any single DeFi venue or bridge implementation.

Control	Objective	Reference behavior
Policy envelope	Bound autonomous actions	Reject moves outside board-approved thresholds
Simulation	Pre-trade validation	Estimate route, slippage, and failure modes
Attestation log	Auditability	Persist model vote, signer approval, and transaction hash

Circuit breakers

LithoVault treats treasury management as a controlled execution problem in which intent, policy, simulation, authorization, and settlement are separate stages that can be logged and reviewed independently. The platform is designed so that AI systems never become unbounded operators; every recommendation is expressed as a typed treasury intent, scored by confidence, constrained by policy, and either executed under delegated limits or escalated to governance. By separating vault state from strategy logic and by separating strategy logic from bridge and protocol adapters, the system can evolve individual components without invalidating accounting or governance assumptions. A treasury deployment can map each business unit, legal entity, or mandate to an isolated vault with its own policy envelope, signer set, asset list, reporting cadence, and permissible protocol adapters. The intended operational benefit is not merely higher yield but tighter control over how liquidity moves, why it moves, and what evidence remains after the fact.

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Control	Objective	Reference behavior
Policy envelope	Bound autonomous actions	Reject moves outside board-approved thresholds
Simulation	Pre-trade validation	Estimate route, slippage, and failure modes
Attestation log	Auditability	Persist model vote, signer approval, and transaction hash

15. Accounting and Reporting

15. Accounting and Reporting presents the reference design for LithoVault within a governed enterprise treasury context. The discussion assumes a deployment on Lithosphere Makalu and uses APP, DNNS, and MultX-compatible interfaces as foundational coordination layers.

P&L

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Attestation log	Auditability	Persist model vote, signer approval, and transaction hash

Board reporting

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Policy envelope	Bound autonomous actions	Reject moves outside board-approved thresholds
Simulation	Pre-trade validation	Estimate route, slippage, and failure modes
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Forensics

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16. API, SDK, and CLI

16. API, SDK, and CLI presents the reference design for LithoVault within a governed enterprise treasury context. The discussion assumes a deployment on Lithosphere Makalu and uses APP, DNNS, and MultX-compatible interfaces as foundational coordination layers.

Interfaces

LithoVault treats treasury management as a controlled execution problem in which intent, policy, simulation, authorization, and settlement are separate stages that can be logged and reviewed independently. The platform is designed so that AI systems never become unbounded operators; every recommendation is expressed as a typed treasury intent, scored by confidence, constrained by policy, and either executed under delegated limits or escalated to governance. By separating vault state from strategy logic and by separating strategy logic from bridge and protocol adapters, the system can evolve individual components without invalidating accounting or governance assumptions. A treasury deployment can map each business unit, legal entity, or mandate to an isolated vault with its own policy envelope, signer set, asset list, reporting cadence, and permissible protocol adapters. The intended operational benefit is not merely higher yield but tighter control over how liquidity moves, why it moves, and what evidence remains after the fact.

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- Policy evaluation precedes execution and determines whether an action can be delegated, queued, rate-limited, or rejected.
- Every material state transition emits a structured event so that operators can reconstruct what happened without depending on off-chain memory.
- Execution adapters must be replaceable and versioned to avoid coupling treasury accounting to any single DeFi venue or bridge implementation.

Control	Objective	Reference behavior
Policy envelope	Bound autonomous actions	Reject moves outside board-approved thresholds
Simulation	Pre-trade validation	Estimate route, slippage, and failure modes
Attestation log	Auditability	Persist model vote, signer approval, and transaction hash

Automation

LithoVault treats treasury management as a controlled execution problem in which intent, policy, simulation, authorization, and settlement are separate stages that can be logged and reviewed independently. The platform is designed so that AI systems never become unbounded operators; every recommendation is expressed as a typed treasury intent, scored by confidence, constrained by policy, and either executed under delegated limits or escalated to governance. By separating vault state from strategy logic and by separating strategy logic from bridge and protocol adapters, the system can evolve individual components without invalidating accounting or governance assumptions. A treasury deployment can map each business unit, legal entity, or mandate to an isolated vault with its own policy envelope, signer set, asset list, reporting cadence, and

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Policy envelope	Bound autonomous actions	Reject moves outside board-approved thresholds
Simulation	Pre-trade validation	Estimate route, slippage, and failure modes
Attestation log	Auditability	Persist model vote, signer approval, and transaction hash

Operator ergonomics

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Policy envelope	Bound autonomous actions	Reject moves outside board-approved thresholds
Simulation	Pre-trade validation	Estimate route, slippage, and failure modes
Attestation log	Auditability	Persist model vote, signer approval, and transaction hash

17. Web and Mobile Experience

17. Web and Mobile Experience presents the reference design for LithoVault within a governed enterprise treasury context. The discussion assumes a deployment on Lithosphere Makalu and uses APP, DNNS, and MultX-compatible interfaces as foundational coordination layers.

Dashboard

LithoVault treats treasury management as a controlled execution problem in which intent, policy, simulation, authorization, and settlement are separate stages that can be logged and reviewed independently. The platform is designed so that AI systems never become unbounded operators; every recommendation is expressed as a typed treasury intent, scored by confidence, constrained by policy, and either executed under delegated limits or escalated to governance. By separating vault state from strategy logic and by separating strategy logic from bridge and protocol adapters, the system can evolve individual components without invalidating accounting or governance assumptions. A treasury deployment can map each business unit, legal entity, or mandate to an isolated vault with its own policy envelope, signer set, asset list, reporting cadence, and permissible protocol adapters. The intended operational benefit is not merely higher yield but tighter control over how liquidity moves, why it moves, and what evidence remains after the fact.

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Simulation	Pre-trade validation	Estimate route, slippage, and failure modes
Attestation log	Auditability	Persist model vote, signer approval, and transaction hash

Approvals

LithoVault treats treasury management as a controlled execution problem in which intent, policy, simulation, authorization, and settlement are separate stages that can be logged and reviewed independently. The platform is designed so that AI systems never become unbounded operators; every recommendation is expressed as a typed treasury intent, scored by confidence, constrained by policy, and either executed under delegated limits or escalated to governance. By separating vault state from strategy logic and by separating strategy logic from bridge and protocol adapters, the system can evolve individual components without invalidating accounting or governance assumptions. A treasury deployment can map each business unit, legal entity, or mandate to an

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Attestation log	Auditability	Persist model vote, signer approval, and transaction hash

Alerts

LithoVault treats treasury management as a controlled execution problem in which intent, policy, simulation, authorization, and settlement are separate stages that can be logged and reviewed independently. The platform is designed so that AI systems never become unbounded operators; every recommendation is expressed as a typed treasury intent, scored by confidence, constrained by policy, and either executed under delegated limits or escalated to governance. By separating vault state from strategy logic and by separating strategy logic from bridge and protocol adapters, the system can evolve individual components without invalidating accounting or governance assumptions. A treasury deployment can map each business unit, legal entity, or mandate to an isolated vault with its own policy envelope, signer set, asset list, reporting cadence, and permissible protocol adapters. The intended operational benefit is not merely higher yield but tighter control over how liquidity moves, why it moves, and what evidence remains after the fact.

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- Policy evaluation precedes execution and determines whether an action can be delegated, queued, rate-limited, or rejected.
- Every material state transition emits a structured event so that operators can reconstruct what happened without depending on off-chain memory.
- Execution adapters must be replaceable and versioned to avoid coupling treasury accounting to any single DeFi venue or bridge implementation.

Control	Objective	Reference behavior
Policy envelope	Bound autonomous actions	Reject moves outside board-approved thresholds
Simulation	Pre-trade validation	Estimate route, slippage, and failure modes
Attestation log	Auditability	Persist model vote, signer approval, and transaction hash

18. Reference Smart Contracts

18. Reference Smart Contracts presents the reference design for LithoVault within a governed enterprise treasury context. The discussion assumes a deployment on Lithosphere Makalu and uses APP, DNNS, and MultX-compatible interfaces as foundational coordination layers.

Vault

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Control	Objective	Reference behavior
Policy envelope	Bound autonomous actions	Reject moves outside board-approved thresholds
Simulation	Pre-trade validation	Estimate route, slippage, and failure modes
Attestation log	Auditability	Persist model vote, signer approval, and transaction hash

Governor

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Policy envelope	Bound autonomous actions	Reject moves outside board-approved thresholds
Simulation	Pre-trade validation	Estimate route, slippage, and failure modes
Attestation log	Auditability	Persist model vote, signer approval, and transaction hash

Registry

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Control	Objective	Reference behavior
Policy envelope	Bound autonomous actions	Reject moves outside board-approved thresholds
Simulation	Pre-trade validation	Estimate route, slippage, and failure modes
Attestation log	Auditability	Persist model vote, signer approval, and transaction hash

19. Deployment Topology

19. Deployment Topology presents the reference design for LithoVault within a governed enterprise treasury context. The discussion assumes a deployment on Lithosphere Makalu and uses APP, DNNS, and MultX-compatible interfaces as foundational coordination layers.

Makalu staging

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Attestation log	Auditability	Persist model vote, signer approval, and transaction hash

Production hardening

LithoVault treats treasury management as a controlled execution problem in which intent, policy, simulation, authorization, and settlement are separate stages that can be logged and reviewed independently. The platform is designed so that AI systems never become unbounded operators; every recommendation is expressed as a typed treasury intent, scored by confidence, constrained by policy, and either executed under delegated limits or escalated to governance. By separating vault state from strategy logic and by separating strategy logic from bridge and protocol adapters, the system can evolve individual components without invalidating accounting or governance assumptions. A treasury deployment can map each business unit, legal entity, or mandate to an isolated vault with its own policy envelope, signer set, asset list, reporting cadence, and

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- Policy evaluation precedes execution and determines whether an action can be delegated, queued, rate-limited, or rejected.
- Every material state transition emits a structured event so that operators can reconstruct what happened without depending on off-chain memory.
- Execution adapters must be replaceable and versioned to avoid coupling treasury accounting to any single DeFi venue or bridge implementation.

Control	Objective	Reference behavior
Policy envelope	Bound autonomous actions	Reject moves outside board-approved thresholds
Simulation	Pre-trade validation	Estimate route, slippage, and failure modes
Attestation log	Auditability	Persist model vote, signer approval, and transaction hash

Ops

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Control	Objective	Reference behavior
Policy envelope	Bound autonomous actions	Reject moves outside board-approved thresholds
Simulation	Pre-trade validation	Estimate route, slippage, and failure modes
Attestation log	Auditability	Persist model vote, signer approval, and transaction hash

20. Security Analysis

20. Security Analysis presents the reference design for LithoVault within a governed enterprise treasury context. The discussion assumes a deployment on Lithosphere Makalu and uses APP, DNNS, and MultX-compatible interfaces as foundational coordination layers.

Threat model

LithoVault treats treasury management as a controlled execution problem in which intent, policy, simulation, authorization, and settlement are separate stages that can be logged and reviewed independently. The platform is designed so that AI systems never become unbounded operators; every recommendation is expressed as a typed treasury intent, scored by confidence, constrained by policy, and either executed under delegated limits or escalated to governance. By separating vault state from strategy logic and by separating strategy logic from bridge and protocol adapters, the system can evolve individual components without invalidating accounting or governance assumptions. A treasury deployment can map each business unit, legal entity, or mandate to an isolated vault with its own policy envelope, signer set, asset list, reporting cadence, and permissible protocol adapters. The intended operational benefit is not merely higher yield but tighter control over how liquidity moves, why it moves, and what evidence remains after the fact.

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Simulation	Pre-trade validation	Estimate route, slippage, and failure modes
Attestation log	Auditability	Persist model vote, signer approval, and transaction hash

Assumptions

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Simulation	Pre-trade validation	Estimate route, slippage, and failure modes
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Residual risks

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Attestation log	Auditability	Persist model vote, signer approval, and transaction hash

21. Economic Model

21. Economic Model presents the reference design for LithoVault within a governed enterprise treasury context. The discussion assumes a deployment on Lithosphere Makalu and uses APP, DNNS, and MultX-compatible interfaces as foundational coordination layers.

Fee flows

LithoVault treats treasury management as a controlled execution problem in which intent, policy, simulation, authorization, and settlement are separate stages that can be logged and reviewed independently. The platform is designed so that AI systems never become unbounded operators; every recommendation is expressed as a typed treasury intent, scored by confidence, constrained by policy, and either executed under delegated limits or escalated to governance. By separating vault state from strategy logic and by separating strategy logic from bridge and protocol adapters, the system can evolve individual components without invalidating accounting or governance assumptions. A treasury deployment can map each business unit, legal entity, or mandate to an isolated vault with its own policy envelope, signer set, asset list, reporting cadence, and permissible protocol adapters. The intended operational benefit is not merely higher yield but tighter control over how liquidity moves, why it moves, and what evidence remains after the fact.

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- Policy evaluation precedes execution and determines whether an action can be delegated, queued, rate-limited, or rejected.
- Every material state transition emits a structured event so that operators can reconstruct what happened without depending on off-chain memory.
- Execution adapters must be replaceable and versioned to avoid coupling treasury accounting to any single DeFi venue or bridge implementation.

Control	Objective	Reference behavior
Policy envelope	Bound autonomous actions	Reject moves outside board-approved thresholds
Simulation	Pre-trade validation	Estimate route, slippage, and failure modes
Attestation log	Auditability	Persist model vote, signer approval, and transaction hash

Inference budgets

LithoVault treats treasury management as a controlled execution problem in which intent, policy, simulation, authorization, and settlement are separate stages that can be logged and reviewed independently. The platform is designed so that AI systems never become unbounded operators; every recommendation is expressed as a typed treasury intent, scored by confidence, constrained by policy, and either executed under delegated limits or escalated to governance. By separating vault state from strategy logic and by separating strategy logic from bridge and protocol adapters, the system can evolve individual components without invalidating accounting or governance assumptions. A treasury deployment can map each business unit, legal entity, or mandate to an isolated vault with its own policy envelope, signer set, asset list, reporting cadence, and

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Control	Objective	Reference behavior
Policy envelope	Bound autonomous actions	Reject moves outside board-approved thresholds
Simulation	Pre-trade validation	Estimate route, slippage, and failure modes
Attestation log	Auditability	Persist model vote, signer approval, and transaction hash

Treasury ROI

LithoVault treats treasury management as a controlled execution problem in which intent, policy, simulation, authorization, and settlement are separate stages that can be logged and reviewed independently. The platform is designed so that AI systems never become unbounded operators; every recommendation is expressed as a typed treasury intent, scored by confidence, constrained by policy, and either executed under delegated limits or escalated to governance. By separating vault state from strategy logic and by separating strategy logic from bridge and protocol adapters, the system can evolve individual components without invalidating accounting or governance assumptions. A treasury deployment can map each business unit, legal entity, or mandate to an isolated vault with its own policy envelope, signer set, asset list, reporting cadence, and permissible protocol adapters. The intended operational benefit is not merely higher yield but tighter control over how liquidity moves, why it moves, and what evidence remains after the fact.

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Policy envelope	Bound autonomous actions	Reject moves outside board-approved thresholds
Simulation	Pre-trade validation	Estimate route, slippage, and failure modes
Attestation log	Auditability	Persist model vote, signer approval, and transaction hash

22. Roadmap

22. Roadmap presents the reference design for LithoVault within a governed enterprise treasury context. The discussion assumes a deployment on Lithosphere Makalu and uses APP, DNNS, and MultX-compatible interfaces as foundational coordination layers.

Near term

LithoVault treats treasury management as a controlled execution problem in which intent, policy, simulation, authorization, and settlement are separate stages that can be logged and reviewed independently. The platform is designed so that AI systems never become unbounded operators; every recommendation is expressed as a typed treasury intent, scored by confidence, constrained by policy, and either executed under delegated limits or escalated to governance. By separating vault state from strategy logic and by separating strategy logic from bridge and protocol adapters, the system can evolve individual components without invalidating accounting or governance assumptions. A treasury deployment can map each business unit, legal entity, or mandate to an isolated vault with its own policy envelope, signer set, asset list, reporting cadence, and permissible protocol adapters. The intended operational benefit is not merely higher yield but tighter control over how liquidity moves, why it moves, and what evidence remains after the fact.

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Simulation	Pre-trade validation	Estimate route, slippage, and failure modes
Attestation log	Auditability	Persist model vote, signer approval, and transaction hash

Medium term

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Attestation log	Auditability	Persist model vote, signer approval, and transaction hash

Research agenda

LithoVault treats treasury management as a controlled execution problem in which intent, policy, simulation, authorization, and settlement are separate stages that can be logged and reviewed independently. The platform is designed so that AI systems never become unbounded operators; every recommendation is expressed as a typed treasury intent, scored by confidence, constrained by policy, and either executed under delegated limits or escalated to governance. By separating vault state from strategy logic and by separating strategy logic from bridge and protocol adapters, the system can evolve individual components without invalidating accounting or governance assumptions. A treasury deployment can map each business unit, legal entity, or mandate to an isolated vault with its own policy envelope, signer set, asset list, reporting cadence, and permissible protocol adapters. The intended operational benefit is not merely higher yield but tighter control over how liquidity moves, why it moves, and what evidence remains after the fact.

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- Policy evaluation precedes execution and determines whether an action can be delegated, queued, rate-limited, or rejected.
- Every material state transition emits a structured event so that operators can reconstruct what happened without depending on off-chain memory.
- Execution adapters must be replaceable and versioned to avoid coupling treasury accounting to any single DeFi venue or bridge implementation.

Control	Objective	Reference behavior
Policy envelope	Bound autonomous actions	Reject moves outside board-approved thresholds
Simulation	Pre-trade validation	Estimate route, slippage, and failure modes
Attestation log	Auditability	Persist model vote, signer approval, and transaction hash

Appendix A. State and Interface Summary

Vault state includes balances, policy identifiers, execution queue pointers, strategy permissions, and accounting snapshots.

Registry state includes DNNS name, controller, zk commitment root reference, and metadata URI.

Bridge intent state includes source vault, destination domain, payload hash, attestation threshold, and settlement receipt.

Appendix B. References

1. MultX Protocol documentation, multx.litho.ai, accessed March 30, 2026.
2. Integrating MultX for Cross-Chain Governance & Treasury Management, multx.litho.ai, accessed March 30, 2026.
3. Lithosphere whitepaper v2.0, paper.litho.ai, accessed March 30, 2026.